

Essam Moussa Mahmoud

Full-Stack Developer – MERN Stack

Alexandria, Egypt | essammoussamahmoud1@gmail.com | +20 102 534 3475 | github.com/essammoussa | [linkedin.com/in/essam-moussa](https://www.linkedin.com/in/essam-moussa) | my-portfolio-essam.vercel.app

SUMMARY

Full-Stack Developer with hands-on MERN experience (MongoDB, Express.js, React.js, Node.js) building role-based, real-time, and API-driven web applications. Implements secure authentication (JWT, OAuth, bcrypt), concurrency-safe operations using MongoDB transactions, and analytics with MongoDB aggregation pipelines. Comfortable owning a feature end-to-end – schema design, REST API, and responsive React UI – and deploying it live.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL, Python, C++
- **Frontend:** React.js, Redux, React Query, Tailwind CSS, Material UI, Shadcn/UI, Responsive Web Design
- **Backend:** Node.js, Express.js, REST API Design, JWT Authentication, OAuth, Socket.IO
- **Databases:** MongoDB, Mongoose, Transactions, Aggregation Pipelines, Database & Schema Design
- **Tools & Practices:** Git, GitHub, Postman, Vercel Deployment, Agile/Scrum Fundamentals

PROJECTS

EasyCare – Healthcare Appointment & Patient Management Platform

2026

React.js · Node.js · Express.js · MongoDB · Mongoose · Socket.IO · JWT · bcrypt · Leaflet · Recharts

- Architected and shipped a full-stack MERN healthcare platform supporting three role-scoped user types (patient, doctor, admin), with ownership-based authorization enforced at the API layer to prevent unauthorized cross-user access.
- Built a concurrency-safe appointment booking engine using MongoDB multi-document transactions, eliminating race-condition double-bookings under simultaneous requests.
- Implemented secure multi-method authentication (JWT, bcrypt password hashing, Google OAuth) with role-based authorization middleware protecting all backend endpoints.
- Delivered real-time patient-doctor messaging via Socket.IO and an admin analytics dashboard (Recharts) powered by custom MongoDB aggregation pipelines for reporting and review moderation.

Wedding Halls Booking Platform

2026

React.js · Node.js · Express.js · MongoDB · REST API

- Delivered a full-stack MERN venue discovery and booking platform as a freelance engagement for an independent client, from requirements gathering through a REST API-driven Express/Node.js backend.
- Designed MongoDB schemas for venues, availability windows, and booking records, exposing RESTful endpoints consumed by the React frontend.
- Developed a responsive React UI for venue discovery, including detail pages with pricing, availability, and image galleries.
- Executed a full visual/UX redesign based on direct client feedback, elevating the platform from a functional MVP to a polished, production-grade product ready for launch.

Library Management System

2025 – 2026

React.js · Redux · React Query · Tailwind CSS · Shadcn/UI · Recharts · Zod

- Engineered a library management system handling book inventory, user accounts, and borrow/return workflows through a reusable, component-driven architecture.
- Implemented schema-validated forms with Zod and a secure authentication flow with sanitized input handling and structured error states.
- Separated global UI state (Redux) from server state (React Query) to enable real-time borrow/return updates without unnecessary re-fetching, improving perceived responsiveness.
- Structured the codebase around reusable, typed components to support long-term maintainability.

React.js · Material UI · Nivo Charts · FullCalendar · Formik · Yup

- Designed a responsive, modular admin dashboard using React and Material UI, prioritizing component reusability across views.
- Implemented interactive data visualizations using Nivo Charts to surface key operational metrics at a glance.
- Integrated calendar-based scheduling and event management with FullCalendar for streamlined operational workflows.
- Built validated, reusable form components with Formik and Yup, reducing invalid-input errors through client-side validation.

EXPERIENCE

Full-Stack Development Intern – TechTrek

Jul 2026 – Present

- Currently completing an intensive full-stack curriculum (React, Next.js, TypeScript, Node.js, Express, MongoDB) spanning frontend fundamentals through backend architecture, database integration, and a final capstone project.
- Building on React core concepts, hooks, state management, and TypeScript in the current phase, with upcoming modules in Node.js, MongoDB, authentication, and deployment.

Full-Stack Web Development Training – Digital Egypt Pioneers Initiative (DEPI)

Nov 2025 – Jul 2026

- Shipped full-stack MERN applications on sprint-style milestones within a structured, project-based curriculum, collaborating with peers in a Git/GitHub team workflow with code review.
- Designed and implemented REST APIs against both SQL and NoSQL databases, delivering end-to-end CRUD and business-logic features under fixed deadlines.
- Architected reusable component libraries and implemented client-side routing and state management across multiple production-style applications.
- Debugged and resolved application-logic defects under deadline pressure to deliver demo-ready features for milestone reviews.

EDUCATION

Alexandria University

B.Sc. Computer and Data Science (In Progress)

Oct 2023 – Expected Jun 2027

CERTIFICATES

Certificate	Issuer	Year
Full-Stack Web Development	Digital Egypt Pioneers Initiative (DEPI)	2026
Developing Back-End Apps with Node.js and Express	Coursera IBM	2026
JavaScript Programming Essentials	Coursera IBM	2025
Developing Front-End Apps with React	Coursera IBM	2025
Getting Started with Git and GitHub	Coursera IBM	2025

LANGUAGES

- Arabic – Native
- English – C1 (Advanced)